

AUTOMATED VEHICLE CLASSIFICATION & COUNTING

AVC&C | Group G25 | University of Bradford

ETHICS & IMPLICATIONS DOCUMENT

CDEI Framework | GDPR | Bias | Privacy | Sustainability | Policy

1. Introduction

This document presents the ethical analysis and broader implications of the AVC&C (Automated Vehicle Classification and Counting) system developed for Bradford Council by Group G25. As an AI system deployed in a public-facing, civic context, the AVC&C system raises important questions around data privacy, algorithmic fairness, public trust, and the responsible deployment of computer vision technology in urban environments.

This analysis is structured around the UK Centre for Data Ethics and Innovation (CDEI) AI Assurance Framework, the UK GDPR and Data Protection Act 2018, and Bradford Council's own commitments to digital inclusion, sustainability (Net Zero 2038), and community trust.

2. CDEI AI Ethics Framework

The Centre for Data Ethics and Innovation (CDEI) provides the UK government's primary framework for responsible AI deployment in public sector contexts. Group G25 assessed the AVC&C system against each of the five CDEI principles:

CDEI Principle	How AVC&C Addresses It	Compliance Status
Safety & Security	Local CPU-only processing. No network transmission of data. No attack surface beyond local device.	COMPLIANT
Transparency	FAIR Report declares all AI tools. Dashboard shows all counts by class and weather. Open-source codebase on GitHub.	COMPLIANT
Fairness	Tested across 5 weather conditions to avoid performance disparities. Vehicle class balance maintained in training data.	COMPLIANT
Accountability	Clear team roles documented. Supervisor oversight at each sprint. Bradford Council client reviews outputs.	COMPLIANT
Contestability	CSV output is auditable. Bradford Council officers can inspect counts. Dashboard provides explainable output.	COMPLIANT

3. GDPR & UK Data Protection Act 2018

3.1 Legal Framework

The AVC&C system operates in public road space where individuals may be incidentally captured on camera. Under the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018, processing of personal data — including images of individuals — requires a lawful basis, explicit justification, and appropriate technical safeguards.

3.2 Lawful Basis Assessment

- **Processing activity:** CCTV-style video capture of public roads for vehicle counting
- **Data controller:** Bradford Council (as the deploying authority in production)
- **Lawful basis:** Article 6(1)(e) — processing necessary for the performance of a task carried out in the public interest (statutory traffic management function)
- **Special category data:** None — the system explicitly filters out facial and biometric data

3.3 Privacy Safeguards Implemented

Safeguard	Implementation	Status
Data Minimisation	Only anonymised vehicle counts stored in CSV. No images, frames, or individuals retained.	IMPLEMENTED
Purpose Limitation	Data collected solely for traffic flow analysis. No secondary use.	IMPLEMENTED
Storage Limitation	No video frames stored post-processing. CSV logs can be deleted per Bradford Council retention policy.	IMPLEMENTED
No Facial Recognition	YOLO class filter restricts detection to vehicles only. No face detection in pipeline.	IMPLEMENTED
No Licence Plate OCR	Bounding box detection only. No character recognition or plate logging.	IMPLEMENTED
Local Processing Only	All inference on local hardware. No cloud upload or external data transfer.	IMPLEMENTED
No Individual Tracking	Track IDs reset each session. No persistent cross-session tracking of specific vehicles.	IMPLEMENTED

The combination of these safeguards means the AVC&C system, as designed, does not constitute processing of personal data within the meaning of the UK GDPR — it processes only anonymised aggregate counts. This represents a privacy-by-design approach aligned with ICO (Information Commissioner's Office) guidance on surveillance systems.

4. Algorithmic Bias Assessment

4.1 Potential Sources of Bias

Any machine learning system trained on historical data carries the risk of encoding biases present in that data. For the AVC&C system, the primary bias risks are:

Bias Type	Description	Mitigation
Geographic Bias	Training data and UA-DETRAC benchmark include Beijing footage, not Bradford-specific road contexts.	Manual validation on Bradford footage. Custom weather dataset sourced specifically for UK conditions.
Vehicle Type Bias	Dataset predominantly contains cars. Buses and trucks are underrepresented in real-world traffic.	Class-balanced sampling in Roboflow. Per-class accuracy validated separately (all >88%).
Weather Bias	Models trained on clear conditions often perform poorly in adverse weather.	Explicitly trained on 5 weather conditions with equal representation in training data.
Exclusion Bias	Motorcycles and cyclists excluded from v1.0 detection. Could under-represent vulnerable road users in count data.	Declared as known limitation. US9 (Could Have) in backlog. Bradford Council informed.
Temporal Bias	System tested on daytime footage only. Night-time or low-light performance unknown.	Declared as known limitation. Night-time testing recommended for v2.0 deployment.

4.2 Bias Mitigation Summary

The most significant unmitigated bias in v1.0 is the exclusion of motorcycles and cyclists. For Bradford Council's traffic planning purposes, this means the system provides an accurate count of motor vehicles but cannot replace comprehensive active travel surveys. This limitation is clearly communicated in all project deliverables and should be addressed before any full production deployment.

5. Privacy Implications

5.1 Public Surveillance Context

Traffic counting cameras operate in public spaces where individuals have a reduced — but not absent — expectation of privacy. The AVC&C system was designed with this tension in mind. Unlike commercial ANPR (Automatic Number Plate Recognition) or facial recognition systems, the AVC&C system processes only vehicle-level bounding box data and discards all frame-level imagery after inference.

5.2 Comparison to Existing Bradford Council Practice

Aspect	Manual Count (Current)	AVC&C System
Data captured	Officer observation — no recording	Video processed locally — no frames stored
Individual identification	Not attempted	Not possible by design
Data retention	Paper tally sheets	Anonymised CSV counts only
Privacy risk level	Very Low	Very Low (equivalent by design)
GDPR implications	None (no personal data)	None (no personal data in output)

The AVC&C system does not increase Bradford Council's privacy risk exposure compared to current manual survey practice. It represents a like-for-like replacement with automation, not an expansion of surveillance capability.

6. Sustainability & Environmental Implications

6.1 Bradford Net Zero 2038

Bradford Council has committed to achieving Net Zero carbon emissions by 2038. Traffic management and modal shift are central to this commitment — reducing private car use requires accurate, real-time data about traffic patterns and trends across Bradford's road network.

The AVC&C system directly supports Bradford's Net Zero 2038 goals by enabling data-driven traffic policy. Specific contributions include:

- **Traffic hotspot identification:** Accurate vehicle counts enable Bradford Council planners to identify congestion points and prioritise active travel infrastructure investment
- **Modal shift monitoring:** Classification by vehicle type (car/bus/truck) enables monitoring of bus usage trends — essential for evaluating public transport policy
- **Reduced officer travel:** Automated counting eliminates the need to deploy 2–3 officers per junction per day, reducing Council vehicle emissions from survey activities

- **Evidence for policy:** Time-series traffic data with weather context provides the evidence base for Low Traffic Neighbourhood (LTN) proposals and 20mph zone decisions

6.2 System Carbon Footprint

Component	Sustainability Note	Assessment
Model training	50 epochs on CPU — ~3 hours. No GPU cluster required. Estimated <0.5 kWh.	LOW IMPACT
Inference (runtime)	CPU-only at 25 FPS — typical laptop power draw ~45W. Very low per-hour energy use.	LOW IMPACT
No cloud services	No AWS/Azure/GCP usage — no data centre energy consumption.	ZERO CLOUD
Hardware reuse	Runs on existing Council hardware — no new equipment procurement required.	SUSTAINABLE
Open source stack	100% open source — no proprietary licensing or vendor lock-in energy overhead.	SUSTAINABLE

7. Social & Policy Implications

7.1 Workforce Implications

The AVC&C system automates a task currently performed by Bradford Council traffic survey officers. This raises workforce implications that Bradford Council should consider in any production deployment:

- **Short-term:** Traffic survey officers are freed from repetitive manual counting — enabling redeployment to higher-value analysis and community engagement roles
- **Skills transition:** Council staff will require basic training to operate the Streamlit dashboard and interpret CSV outputs — achievable with minimal investment
- **Job displacement risk:** Manual counting is a small part of traffic officer roles. Displacement risk is LOW — the system augments rather than replaces the analytical workforce
- **Recommendation:** Bradford Council should engage affected staff early in any deployment plan and frame the technology as a tool that enhances their role

7.2 Public Trust & Community Implications

Bradford is a diverse city with communities that have historically been subject to disproportionate surveillance. Deployment of any camera-based AI system in Bradford requires particular sensitivity to these dynamics.

- **Communication:** Bradford Council should publish a clear public notice explaining the purpose of AVC&C cameras, what data is collected, and what it is not (no facial recognition, no plate reading)
- **Community engagement:** Consultation with Bradford's communities — particularly in areas where cameras are first deployed — is recommended to build trust
- **Scope discipline:** The AVC&C system must not be repurposed for enforcement or surveillance without a separate legal basis and public consultation
- **Transparency portal:** Making aggregate count data publicly available (e.g. on Bradford Council Open Data portal) would reinforce transparency and public benefit

7.3 Policy Alignment

Policy / Strategy	Relevance to AVC&C	Alignment
Bradford Net Zero 2038	Traffic data enables evidence-based active travel investment	ALIGNED
UK National AI Strategy 2021	Responsible AI deployment in public sector — local government use case	ALIGNED
UK GDPR / DPA 2018	Privacy-by-design, data minimisation, no personal data in output	COMPLIANT
CDEI AI Assurance Framework	All 5 principles addressed (Section 2 above)	COMPLIANT

Bradford District Local Plan	Traffic data supports sustainable transport infrastructure planning	SUPPORTIVE
Highway Code (UK)	System counts road users — does not enforce or interact with live traffic	NO CONFLICT

8. Recommendations for Full Deployment

Based on this ethics and implications assessment, Group G25 makes the following recommendations to Bradford Council before any full production deployment of the AVC&C system:

#	Recommendation	Rationale	Priority
1	Conduct Data Protection Impact Assessment (DPIA)	Required under UK GDPR Article 35 for systematic monitoring of public spaces	HIGH
2	Add motorcycle and bicycle detection (v2.0)	Current exclusion creates an incomplete picture of vulnerable road users	MEDIUM
3	Test on Bradford CCTV footage at scale	Current validation is limited to one Bradford test clip (48 vehicles)	HIGH
4	Publish public-facing transparency notice	Addresses community trust concerns — especially important in Bradford context	HIGH
5	Deploy on NVIDIA Jetson for edge processing	Eliminates need to send video to central server — enhances privacy and reduces latency	MEDIUM
6	Integrate with Bradford Council Open Data portal	Supports transparency and enables wider public benefit from traffic data	MEDIUM
7	Night-time and low-light performance testing	Current system untested in dark conditions — essential for year-round UK deployment	MEDIUM

9. Conclusion

The AVC&C system represents a responsible, privacy-preserving application of AI in a public sector context. It addresses a genuine operational need for Bradford Council, replaces an expensive and inconsistent manual process, and does so without creating new privacy risks or surveillance capabilities.

The system aligns with the CDEI AI Assurance Framework, complies with UK GDPR and the Data Protection Act 2018, and supports Bradford Council's Net Zero 2038 and sustainable transport commitments. The primary ethical gap — the exclusion of motorcycles and cyclists — is acknowledged and documented as a v2.0 development priority.

With appropriate community engagement, a formal DPIA, and the scale testing recommended above, the AVC&C system is suitable for phased production deployment across Bradford's road network.

Document prepared: April 17, 2026 | Group G25 Submission — COS6032-E (Canvas + GitHub)